

Algebra: Chapter 0 Solutions

Mango's Reading Group

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Chapter I

Preliminaries: Set theory and categories

1 Naive Set theory

By: quickdooom

Ex 1.1. Russell's Paradox shows that naïve set theory's assumption of the unrestricted comprehension principle – for any property P , the set $\{x : P(x)\}$ exists – leads to a contradiction. Consider the property $P(x) : x \in x$. The unrestricted comprehension principle tells us that the set $S = \{x : \sim P(x)\}$ exists. This is simply not possible, since we get $S \in S \iff S \notin S$.

To see this, note that if S contains itself, then S does not satisfy $\sim P(S)$, so $S \notin S$, but now S does satisfy $\sim P(S)$, so $S \in S$. i.e, $S \in S \iff S \notin S$, and we know that only one of those alternatives is possible. A more detailed discussion can be found at <https://plato.stanford.edu/archives/fall2025/entries/russell-paradox/>

Ex 1.2. First, we check that the elements of $\mathcal{P}_\sim$ are nonempty. Indeed, since $\sim$ is an equivalence relation, we have that for all $a \in S : a \sim a$, so that $a \in [a]$ i.e $[a]$ is nonempty. Now we check that distinct elements of $\mathcal{P}_\sim$ are disjoint. Indeed, let $[a] \neq [b]$ be two elements of $\mathcal{P}_\sim$. If they did intersect at some c , we would have $a \sim c$ and $b \sim c$ which by transitivity would give $a \sim b$. This is simply not possible, because this implies by transitivity that for every $x \in S$ such that $x \sim b$, we have $x \sim a$. i.e $[b] \subseteq [a]$. Symmetry gives $[a] \subseteq [b]$ i.e $[a] = [b]$, which contradicts our initial assumption of $[a] \neq [b]$. Ofcourse the union of the elements of $\mathcal{P}_\sim$ is S because the elements of $\mathcal{P}_\sim$ are $[a] \supseteq \{a\}$ for each $a \in S$. This gives us

$$\bigcup_{a \in S} [a] \supseteq S.$$

The other inclusion is clear, since each $[a]$ is a subset of S . Thus we have $\bigcup_{x \in \mathcal{P}_\sim} x = \bigcup_{a \in S} [a] = S$ as desired. i.e, $\mathcal{P}_\sim$ is a partition of S .

Ex 1.3. let $\mathcal{P} = \{S_\alpha : \alpha \in I\}$ be a partition of S . We claim that the equivalence relation $\sim$ given by $a \sim b \iff (\exists \alpha \in I : a \in S_\alpha \text{ and } b \in S_\alpha)$. I.e two elements are related iff they lie in the same part of the partition. Indeed, this is an equivalence relation. For any $a \in S$, we have $a \sim a$ because $a \in S_\alpha$ for some α so $\sim$ is reflexive. Now suppose that $a \sim b$ and $b \sim c$. Since b lies in exactly one S_α , we have that a and c lie in the same S_α , i.e $a \sim c$ so $\sim$ is transitive. Symmetry is clear, if a lies in the same part as b , b lies in the same part as a .

Ex 1.4. From what we have seen, it suffices to count the number of partitions of $\{1, 2, 3\}$, which are $\{\{1\}, \{2\}, \{3\}\}, \{\{1, 2\}, \{3\}\}, \{\{1, 3\}, \{2\}\}, \{\{2, 3\}, \{1\}\}, \{\{1, 2, 3\}\}$. I.e, there are 5 equivalence relations

on the set $\{1, 2, 3\}$

Ex 1.5. On the set $\{1, 2, 3\}$ define the following equivalence relation:

$$1 \sim 1, 2 \sim 2, 3 \sim 3, 3 \sim 2, 2 \sim 1, 2 \sim 3, 1 \sim 2.$$

This is reflexive and symmetric but not transitive. Indeed, $1 \sim 2$ and $2 \sim 3$ but $1 \not\sim 3$. If we use a relation that is only reflexive and symmetric but not transitive, then we will not get a partition. Indeed, the parts may intersect. For example, in the above relation we defined, if we let $[a] := \{b : a \sim b\}$, then $[1] \cap [2] = \{2\}$ even though $[1] \neq [2]$.

Ex 1.6. The relation $\sim$ is reflexive since indeed, $a - a = 0 \in \mathbb{Z}$ for all $a \in \mathbb{R}$. It is symmetric, since the negative of an integer is an integer. Indeed, $a \sim b \implies a - b = k_1 \implies b - a = -k_1$. It is transitive, since the sum of two integers is an integer. Indeed, if $a \sim b$, $b \sim c$, we have $a - b = k_1$, $b - c = k_2$ for some $k_1, k_2 \in \mathbb{Z}$ which gives $a - c = k_1 + k_2 \in \mathbb{Z}$. Hence, it is an equivalence relation. In exactly the same way, we get that the relation $(a_1, a_2) \approx (b_1, b_2) \iff a_1 - b_1 \in \mathbb{Z}$ and $a_2 - b_2 \in \mathbb{Z}$ is an equivalence relation.

As for finding “compelling” descriptions, first relation can be seen geometrically as a ‘circle’ in the sense that the set $\mathbb{R}/\sim$ can be identified with the set $\{e^{2\pi ix}\}$ via $[x] \mapsto e^{2\pi ix}$. It is important to note that since we are using the representative of the classes in defining our correspondence, that it is well defined. Similarly, the set $\mathbb{R} \times \mathbb{R}/\approx$ can be identified with a torus, via the same exponential map: $[(x, y)] \mapsto (e^{2\pi ix}, e^{2\pi iy})$ which one can again check is well defined.

2 Functions Between Sets

By: blanketism

Ex 2.1. The number of bijections from a set of size n to itself is $n!$. Indeed, we can choose the image of the first element in n ways, the image of the second element in $n - 1$ ways, and so on. This gives us $n(n - 1)(n - 2) \cdots 1 = n!$ bijections.

Ex 2.2. Let $f : A \rightarrow B$ be a function with $A \neq \emptyset$. Suppose that f has a right-inverse $g : B \rightarrow A$ so that $f \circ g = \text{id}_B$. We claim that f is surjective. Indeed, let $b \in B$. Then we have $f(g(b)) = b$ so that b is in the image of f . Since b was arbitrary, we have that f is surjective.

Now suppose that f is surjective. We claim that f has a right-inverse. By surjectivity of f , for all $b \in B$, there exists $a_b \in A$ such that $f(a_b) = b$. Define $g : B \rightarrow A$ by $g(b) = a_b$. Then we have $f(g(b)) = f(a_b) = b$ so that $f \circ g = \text{id}_B$. Hence, f has a right-inverse. (Note that our choice of a_b is dependent on the axiom of choice.)

Ex 2.3. Let $f : A \rightarrow B$ be a bijection with inverse $f^{-1} : B \rightarrow A$. We claim that f^{-1} is a bijection. Indeed, since f is surjective, for all $a \in A$, there exists $b \in B$ such that $f(a) = b$. Then we have $f^{-1}(b) = f^{-1}(f(a)) = a$ so that f^{-1} is surjective. Now suppose that $f^{-1}(b_1) = f^{-1}(b_2)$ for some $b_1, b_2 \in B$. Then we have $f(f^{-1}(b_1)) = f(f^{-1}(b_2))$ which gives us $b_1 = b_2$ so that f^{-1} is injective. Hence, f^{-1} is a bijection.

Now suppose that $f : A \rightarrow B$ and $g : B \rightarrow C$ are bijections. We claim that $g \circ f : A \rightarrow C$ is a bijection. Indeed, since f is surjective, for all $c \in C$, there exists $b \in B$ such that $g(b) = c$. Since g is surjective, there exists $a \in A$ such that $f(a) = b$. Then we have $(g \circ f)(a) = g(f(a)) = g(b) = c$ so that $g \circ f$ is surjective. Now suppose that $(g \circ f)(a_1) = (g \circ f)(a_2)$ for some $a_1, a_2 \in A$. Then we have $g(f(a_1)) = g(f(a_2))$. Since g is injective, we have $f(a_1) = f(a_2)$. Since f is injective, we have $a_1 = a_2$. Hence, $g \circ f$ is injective. Thus, we have shown that the composition of two bijections is a bijection.

Ex 2.4. We claim that the notion of isomorphism is an equivalence relation. Indeed, let A, B, C be sets. Then we have the following:

- Reflexivity: The identity map $\text{id}_A : A \rightarrow A$ is a bijection, so A is isomorphic to itself.
- Symmetry: If $f : A \rightarrow B$ is a bijection, then its inverse $f^{-1} : B \rightarrow A$ is also a bijection, so if A is isomorphic to B , then B is isomorphic to A .
- Transitivity: If $f : A \rightarrow B$ and $g : B \rightarrow C$ are bijections, then their composition $g \circ f : A \rightarrow C$ is also a bijection, so if A is isomorphic to B and B is isomorphic to C , then A is isomorphic to C .

Ex 2.5. Analogous to a monomorphism, we can formulate the notion of an epimorphism as follows: A function $f : A \rightarrow B$ is an epimorphism (or epic), if the following holds:

$$\begin{aligned} &\text{for all sets } Z \text{ and all functions } g_1, g_2 : B \rightarrow Z, \\ &g_1 \circ f = g_2 \circ f \implies g_1 = g_2. \end{aligned}$$

We now prove a result analogous to Proposition 2.3 for epimorphisms: A function is surjective if and only if it is an epimorphism. Indeed, let $f : A \rightarrow B$ be surjective. We claim that f is an epimorphism. Let $g_1, g_2 : B \rightarrow Z$ be functions such that $g_1 \circ f = g_2 \circ f$. We want to show that $g_1 = g_2$. Since f is surjective, it has a right-inverse $h : B \rightarrow A$ such that $f \circ h = \text{id}_B$. Then we have

$$g_1 \circ (f \circ h) = (g_1 \circ f) \circ h = (g_2 \circ f) \circ h = g_2 \circ (f \circ h)$$

Since h is a right inverse of f , we have $f \circ h = \text{id}_B$, so we get $g_1 \circ \text{id}_B = g_2 \circ \text{id}_B$, which gives us $g_1 = g_2$. Hence, f is an epimorphism.

Now suppose that $f : A \rightarrow B$ is an epimorphism. We claim that f is surjective. Suppose not, then there exists $b_0 \in B$ such that $b_0 \notin f(A)$. Define two functions $g_1, g_2 : B \rightarrow \{0, 1\}$ by

$$g_1(b) = \begin{cases} 0 & \text{if } b = b_0 \\ 1 & \text{otherwise} \end{cases} \quad \text{and} \quad g_2(b) = 1 \text{ for all } b \in B.$$

Then we have $g_1 \circ f = g_2 \circ f$, since $b_0 \notin f(A)$, but $g_1 \neq g_2$, which contradicts the assumption that f is an epimorphism. Hence, f must be surjective.

Ex 2.6. Recall that a section of a surjection $f : A \rightarrow B$ is a right-inverse of f , i.e a function $g : B \rightarrow A$ such that $f \circ g = \text{id}_B$. Given $f : A \rightarrow B$, define the map $s : A \rightarrow A \times B$ as

$$s(a) = (a, f(a)).$$

Then for all $a \in A$, we have $\pi_A(s(a)) = \pi_A(a, f(a)) = a$, so that $\pi_A \circ s = \text{id}_A$. Hence, s is a section of π_A .

Conversely, given a section $s : A \rightarrow A \times B$ of π_A , we can define a function $f : A \rightarrow B$ by $f(a) = \pi_B(s(a))$. Then for all $a \in A$, we have $\pi_B(s(a)) = f(a)$, so that $s(a) = (a, f(a))$. Hence, we can recover the original function f from the section s .

Ex 2.7. Let $f : A \rightarrow B$ be a function with graph Γ_f . To see that Γ_f is isomorphic to A , we can define a function $g : A \rightarrow \Gamma_f$ by $g(a) = (a, f(a))$. Then for all $a_1, a_2 \in A$, we have

$$g(a_1) = g(a_2) \iff (a_1, f(a_1)) = (a_2, f(a_2)) \iff a_1 = a_2 \text{ and } f(a_1) = f(a_2) \iff a_1 = a_2,$$

so that g is injective. Now let $(a, b) \in \Gamma_f$. Then we have $b = f(a)$, so that $g(a) = (a, f(a)) = (a, b)$. Hence, g is surjective. Therefore, g is a bijection, and we have shown that Γ_f is isomorphic to A .

Ex 2.8. Let $f : \mathbb{R} \rightarrow \mathbb{C}$ be defined by $f : r \mapsto e^{2\pi i r}$. Then the canonical decomposition of f is given by the following diagram:

$$\begin{array}{ccccccc} & & & f & & & \\ & & \nearrow & & \searrow & & \\ \mathbb{R} & \longrightarrow & \mathbb{R}/\sim & \xrightarrow[\tilde{f}]{\sim} & \text{im } f & \hookrightarrow & \mathbb{C} \end{array}$$

We now describe all the terms in the canonical decomposition:

- The equivalence relation $\sim$ is defined by $r_1 \sim r_2 \iff f(r_1) = f(r_2) \iff e^{2\pi i r_1} = e^{2\pi i r_2} \iff r_1 - r_2 \in \mathbb{Z}$. Hence, the equivalence classes are given by $[r] = \{r + n : n \in \mathbb{Z}\}$ for each $r \in \mathbb{R}$.
- The quotient set $\mathbb{R}/\sim$ is the set of all equivalence classes, which can be identified with the interval $[0, 1)$ by mapping each equivalence class $[r]$ to its unique representative in $[0, 1)$.
- The map $\tilde{f} : \mathbb{R}/\sim \rightarrow \text{im } f$ is defined by $\tilde{f}([r]) = f(r) = e^{2\pi i r}$. This map is well-defined, since if $[r_1] = [r_2]$, then $r_1 - r_2 \in \mathbb{Z}$, and hence $e^{2\pi i r_1} = e^{2\pi i r_2}$.
- The image $\text{im } f$ is the set of all complex numbers of the form $e^{2\pi i r}$ for $r \in \mathbb{R}$, which is the unit circle in the complex plane, denoted by $S^1 = \{z \in \mathbb{C} : |z| = 1\}$.
- The inclusion map $\text{im } f \hookrightarrow \mathbb{C}$ is the natural inclusion of the unit circle into the complex plane.

Moreover, this exercise is exactly the same as Exercise 1.6.

Ex 2.9. Define sets A', A'', B', B'' such that $A' \cong A'', B' \cong B''$, and $A' \cap B' = A'' \cap B'' = \emptyset$. We claim that $f : A' \cup B' \rightarrow A'' \cup B''$ defined by

$$f(x) = \begin{cases} f_1(x) & \text{if } x \in A' \\ f_2(x) & \text{if } x \in B' \end{cases}$$

is a bijection, where $f_1 : A' \rightarrow A''$ and $f_2 : B' \rightarrow B''$ are bijections. Indeed, let $x_1, x_2 \in A' \cup B'$ such that $f(x_1) = f(x_2)$. We have the following cases:

- If $x_1, x_2 \in A'$, then $f_1(x_1) = f_1(x_2)$, and since f_1 is a bijection, we have $x_1 = x_2$.
- If $x_1, x_2 \in B'$, then $f_2(x_1) = f_2(x_2)$, and since f_2 is a bijection, we have $x_1 = x_2$.

Therefore, f is injective. Now let $y \in A'' \cup B''$. We have the following cases:

- If $y \in A''$, then there exists $x \in A'$ such that $f_1(x) = y$, and hence $f(x) = y$.
- If $y \in B''$, then there exists $x \in B'$ such that $f_2(x) = y$, and hence $f(x) = y$.

Therefore, f is surjective, and hence a bijection. One should note that the cases of x_1, x_2 and y being in different sets are impossible, since $A' \cap B' = A'' \cap B'' = \emptyset$.

From this, we see that the operation $A \amalg B$ is well-defined up to isomorphism, since if $A' \cong A''$ and $B' \cong B''$, then $A' \amalg B' \cong A'' \amalg B''$.

Ex 2.10. Let A and B be sets. To see that $|B^A| = |B|^{|A|}$, let $|A| = m$ and $|B| = n$. Then we have that B^A is the set of all functions from A to B . For each element of A , there are n choices for its image in B . Since there are m elements in A , the total number of functions from A to B is given by the product of the number of choices for each element, which is

$$|B^A| = n^m = |B|^{|A|}.$$

Ex 2.11. Let 2^A denote the set of functions from A to $\{0, 1\}$. We claim that $2^A \cong \mathcal{P}(A)$. Indeed, we can define a bijection $f : 2^A \rightarrow \mathcal{P}(A)$ by

$$f(g) = \{a \in A : g(a) = 1\}.$$

To see that f is injective, let $g_1, g_2 \in 2^A$ such that $f(g_1) = f(g_2)$. Then we have

$$\{a \in A : g_1(a) = 1\} = \{a \in A : g_2(a) = 1\}.$$

This implies that for all $a \in A$, $g_1(a) = 1$ if and only if $g_2(a) = 1$, and hence $g_1 = g_2$. Therefore, f is injective. To see that f is surjective, let $S \in \mathcal{P}(A)$. We can define a function $g : A \rightarrow \{0, 1\}$ by

$$g(a) = \begin{cases} 1 & \text{if } a \in S \\ 0 & \text{if } a \notin S \end{cases}$$

Then we have $f(g) = S$, and hence f is surjective.